Welcome to Introduction to Web Scraping!

Please log onto the Etherpad:
https://pad.carpentries.org/2026-03-10-bham-intro-web-scraping

Add your details in the ‘Sign in’ section

Check if you have completed the Software Setup’

html code from Canadian MPs webpage

```
(...)
<tr role="row" id="mp-list-id-25446">
    <td data-sort="Allison Dean" class="sorting_1">
        <a href="/members/en/dean-allison(25446)">
            Allison, Dean
        </a>
    </td>
    <td data-sort="Conservative">Conservative</td>
    <td data-sort="Niagara West">
        <a href="/members/en/constituencies/niagara-west(1124)">Niagara West</a>
    </td>
    <td data-sort="Ontario">Ontario</td>
</tr>
```

html code from UK MPs webpage

```
(...)
<a class="card card-member" href="/member/172/contact">
    <div class="card-inner">
        <div class="content">
            <div class="image-outer">
                <div class="image"
                    aria-label="Image of Ms Diane Abbott"
                    style="background-image: url(https://members-api.parliament.uk/api/Members/172/Thumbnail); border-color: #909090;"></div>
            </div>
            <div class="primary-info">
                Ms Diane Abbott
            </div>
            <div class="secondary-info">
                Independent
            </div>
        </div>
        <div class="info">
            <div class="info-inner">
                <div class="indicators-left">
                    <div class="indicator indicator-label">
                        Hackney North and Stoke Newington
                    </div>
                </div>

                <div class="clearfix"></div>
            </div>
        </div>
    </div>
</a>
(...)
```

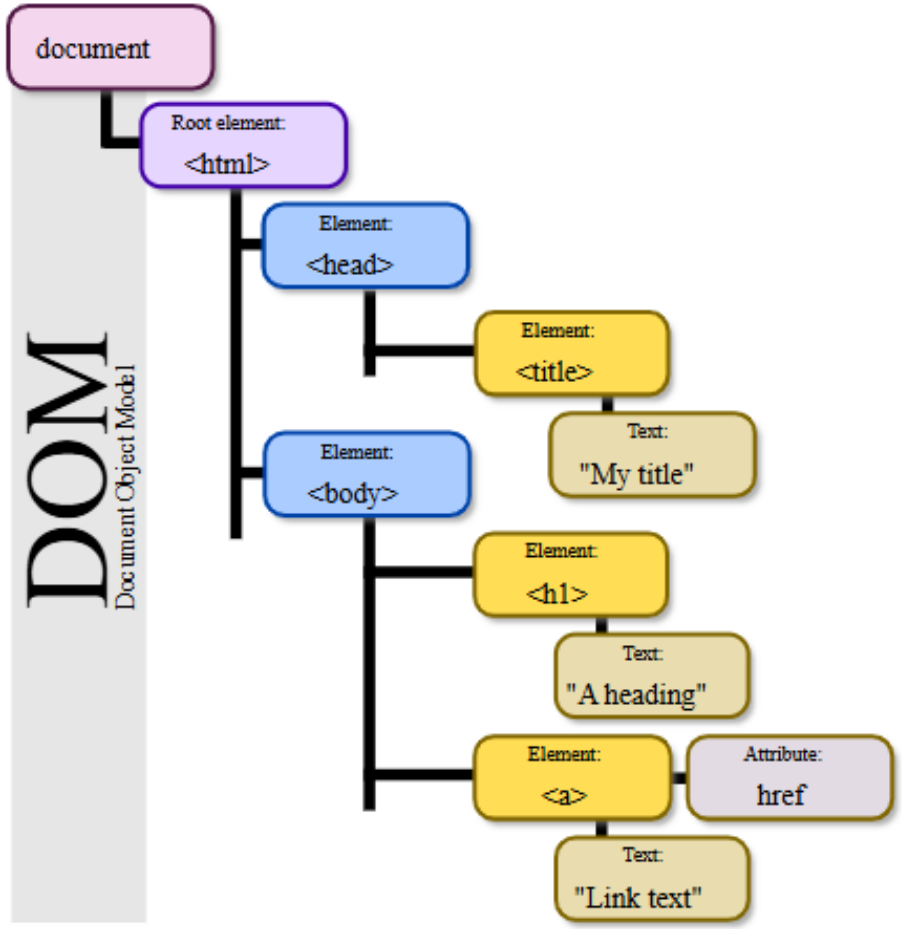
"Web scraping (web harvesting or web data extraction) is a computer software technique of extracting information from websites."
Wikipedia https://en.wikipedia.org/wiki/Web_scraping

Web scraping typically targets one web site at a time to extract unstructured information and put it in a structured form for reuse.

Hypertext Markup Language – html

```html
<!DOCTYPE html>
<html>
<head>
  <title>Sample web page</title>
</head>
<body>
  <h1>h1 Header #1</h1>
  <p>This is a paragraph tag</p>
  <h2>h2 Sub-header</h2>
  <p>A new paragraph, now in the <b>sub-header</b></p>
    <h1>h1 Header #2</h1>
    <p>
    This other paragraph has two hyperlinks,
    one to <a href="https://carpentries.org/">The Carpentries homepage</a>,
    and another to the
    <a href="https://carpentries.org/workshops/past-workshops/">past workshops</a>
  </p>
</body>
</html>
```

html Tree Format



Source: Wikipedia. Author: Birger Eriksson

Cascading Style Sheet (CSS) Code

```
<head>
  <title>Sample web page</title>
  <style>
        /* CSS Rule */
        h1 {
            color: blue;
            /* Property: value */
            font-size: 24px;
        }

        p {
            color: green;
            font-size: 16px;
        }
    </style>
</head>
```

Email address

```
(...)
<div class="col-md-7">
   <div class="contact-line">
      <span class="label">Phone Number:</span>
      <a href="tel:020 7219 4426">020 7219 4426</a>
   </div>
   <div class="contact-line">
      <span class="label">Email Address:</span>
      <a href="mailto:diane.abbott.office@parliament.uk"> diane.abbott.office@parliament.uk</a>
   </div>
</div>
(...)
```

Selector: div.contact-line:nth-of-type(2) a

2026 IT Needs of Active Research Survey

- Survey going live Monday 23 February for 4 weeks
  - Being sent to all research staff via new mailing list
  - PGR Leads being asked to send out to PGRs
- Your opportunity to shape the future of research computing at UoB!
  - Received 5 years of funding for extra training based on 2024 results
- Survey link
  - https://app.onlinesurveys.jisc.ac.uk/s/bham/2026-it-needs-of-active-research

SHAPE THE FUTURE!

SURVEY OPEN UNTIL 23 MARCH

UNIVERSITY OF BIRMINGHAM